THESIS PROGRESS · ESS IN DAY-AHEAD MARKETS

From dual-based to price-based ESS bids

Writing the equivalent bids with nodal prices and private data only,
and what the non-uniqueness of π implies for them



Carlos · October 2026

Where we left off: bids built from duals

RECAP

Equivalent bids (E)

$$u^{p\star}_{e,t} = \alpha_e\left(\pi^\star_{e,t} - \bar{\nu}^\star_{e,t}\right) - \epsilon_e, \quad \hat{p}_{e,t} = \bar{p}_e \qquad c^{g\star}_{e,t} = \zeta_e + \beta_e\left(\pi^\star_{e,t} + \underline{\nu}^\star_{e,t}\right), \quad \hat{g}_{e,t} = \bar{g}_e$$

Corollary 1: unrolling π★

$$c^{g\star}_{e,t} = \zeta_e + \beta_e\left[\gamma_e^{T-t}\,\pi^\star_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}\left(\underline{\nu}^\star_{e,\tau} - \bar{\nu}^\star_{e,\tau}\right) + \underline{\nu}^\star_{e,t}\right]$$

$$u^{p\star}_{e,t} = \alpha_e\left[\gamma_e^{T-t}\,\pi^\star_{e,T} + \sum_{\tau=t+1}^{T} \gamma_e^{\tau-t}\left(\underline{\nu}^\star_{e,\tau} - \bar{\nu}^\star_{e,\tau}\right) - \bar{\nu}^\star_{e,t}\right] - \epsilon_e$$

Problem

π★ and v★ are outputs of Baldick's clearing. In the current design nobody computes them.

Goal

Rewrite (E) with private data and nodal prices only, plus the ESS's own schedule.

Three steps

1. Remove the terminal value and the rents: bids depend on π only
2. Pin π with the prices at the marginal intervals
3. Choose π where the prices leave it free

Prop. 1: same KKT ⇒ same dispatch and nodal prices. Notation and (A)/(S) numbering as in the previous deck and the equivalence note.

Bids depend only on the value of stored energy

STEP 1 · ELIMINATE THE DUALS

Terminal value

$$\pi^{\star}_{e,T} = w_e \qquad \text{(S7), linear terminal bid}$$

Net SOC rent: the jump of π across interval t

$$r_{e,t} := \underline{\nu}_{e,t} - \bar{\nu}_{e,t} = \frac{\pi_{e,t-1}}{\gamma_e} - \pi_{e,t}$$

Assumption 1: at most one SOC limit active per interval ⇒

$$\underline{\nu}_{e,t} = \left[r_{e,t}\right]^{+} \text{(empty)}, \qquad \bar{\nu}_{e,t} = \left[-r_{e,t}\right]^{+} \text{(full)}$$

Corollary 3: substitute into (E)

$$c^{g}_{e,t} = \zeta_e + \beta_e \max\left\{\pi_{e,t},\ \frac{\pi_{e,t-1}}{\gamma_e}\right\}$$

$$u^{p}_{e,t} = \alpha_e \min\left\{\pi_{e,t},\ \frac{\pi_{e,t-1}}{\gamma_e}\right\} - \epsilon_e$$

Reading

- Interval t only needs the value of stored energy at its start and at its end.
- The offer uses the higher value, the bid the lower one.
- The bid–offer gap widens only when the SOC hits a limit (nonzero rent).

Still a dual

The value of stored energy π itself. Steps 2–3 write it with prices.

Three objects that let prices replace the duals

STEP 2 · SEGMENTS, MARGINAL INTERVALS AND VALUES

Baldick base case (solver duals)

t = 1	t = 2 · hits empty	t = 3	t = 4 · hits full	t = 5	t = 6 · hits empty
charge 100 MW (T)	discharge 90 MW	charge 200 MW (T)	charge 22.2 MW	discharge 175 MW (T)	discharge 25 MW
λ★ = 43.1 · SOC 360	λ★ = 50 · SOC 0	λ★ = 0 · SOC 720	λ★ = 0 · SOC 800	λ★ = 50 · SOC 100	λ★ = 50 · SOC 0
$S_1 = \{1\}$: τ = 1, v = 49	$S_2 = \{2, 3\}$: τ = 3, v = 1.11		$S_3 = \{4, 5\}$: τ = 5, v = 49		$S_4 = \{6\}$, last: v = w = 40

Segment S

- Consecutive hours from one limit-hit hour up to the hour before the next one.
- No SOC rent inside, so π only changes by retention: π(t) = γ·π(t+1). With γ = 1, π is constant.

Marginal interval τ

- An hour of the segment where the ESS charges or discharges strictly between 0 and its power limit, and that action's SOC limit is not binding.
- τ(t): the marginal interval of the segment containing t. Last segment: τ := T.

Value v of the segment

At τ the ESS is indifferent at the price, so the price reveals what 1 MWh stored is worth:

$$v = \begin{cases} \dfrac{\lambda_\tau - \zeta}{\beta} & \text{ESS discharging at } \tau \\ \dfrac{\lambda_\tau + \epsilon}{\alpha} & \text{ESS charging at } \tau \\ w & \text{last segment} \end{cases}$$

t = 2 also discharges partially, but it empties the battery: its price does not reveal π. That is why a limit-hit hour is not marginal for the action that hits the limit.

Equivalent bids without dual variables

STEP 3 · PRICE-BASED BIDS

Proposition 4: value of stored energy from prices

$$\pi_{e,t} = \begin{cases} \gamma_e^{\tau_t - t}\, v_{\tau_t}(\lambda), & t \text{ in a segment with marginal interval } \tau_t \\ \gamma_e^{T-t}\, w_e, & t \text{ in the last segment} \end{cases}$$

- The value of 1 MWh stored at the end of t is the value v of its segment.
- With γ < 1 it is discounted once per hour between t and τ(t).

Corollary 4: Corollary 3 with π from Proposition 4

$$c^g_{e,t} = \zeta_e + \beta_e \max\left\{\gamma_e^{\tau_t - t}\, v_{\tau_t},\ \gamma_e^{\tau_{t-1} - t}\, v_{\tau_{t-1}}\right\}$$

$$u^p_{e,t} = \alpha_e \min\left\{\gamma_e^{\tau_t - t}\, v_{\tau_t},\ \gamma_e^{\tau_{t-1} - t}\, v_{\tau_{t-1}}\right\} - \epsilon_e$$

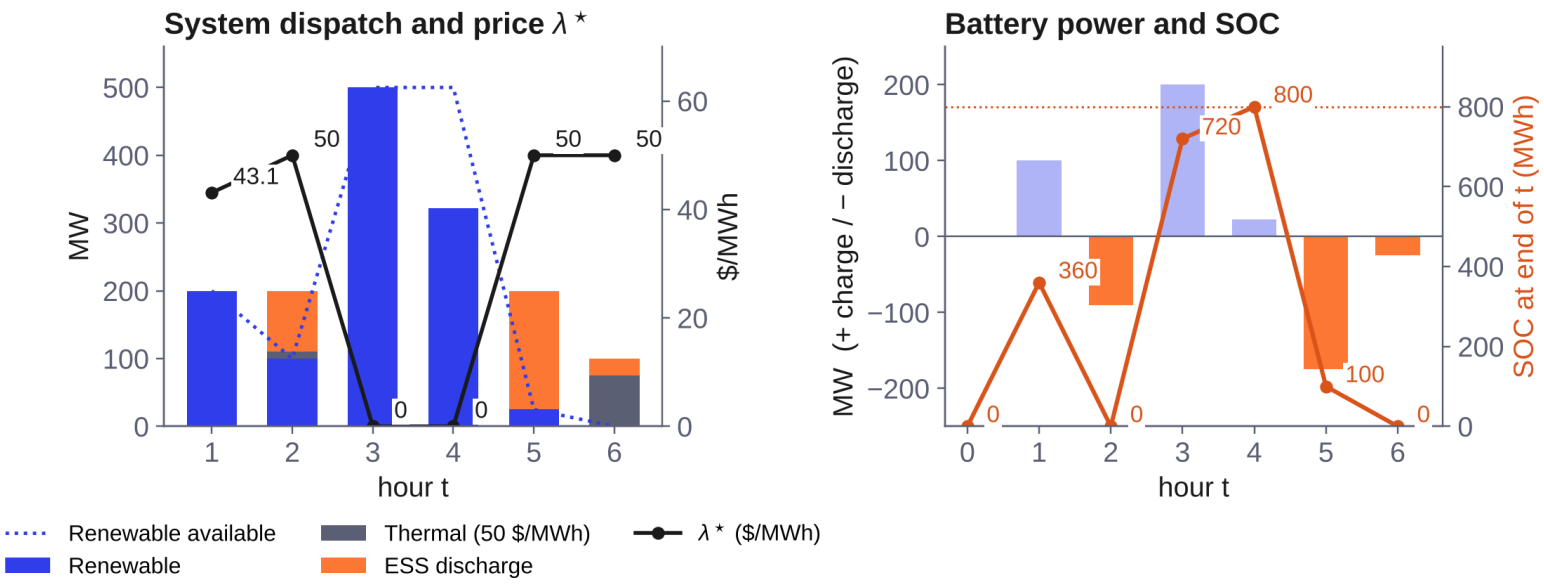
- Hour t compares the value at its end (segment of t) with the value at its start (segment of t−1).
- Same segment: max = min, narrow band. New segment: the band widens.

Base case, solver duals	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
τ(t) → value at end of t	1 → 49	3 → 1.11	3 → 1.11	5 → 49	5 → 49	T → 40
τ(t−1) → value at start of t	(1) → 49	1 → 49	3 → 1.11	3 → 1.11	5 → 49	5 → 49
Offer c^g = 1 + max	50	**50**	2.11	**50**	50	**50**
Bid u^p = 0.9·min − 1	43.1	**0**	0	**0**	43.1	**35**

Teal: hours where t and t−1 lie in different segments (band widens). Inputs: private data, prices at the marginal intervals, and the hours that hit a limit, all from the ESS's own self-schedule.

The price-based formulas recover the solver's multipliers

WORKED EXAMPLE · BALDICK BASE CASE



Single bus, T = 6, ΔT = 4 h. Renewable at 0 $/MWh, thermal at 50 $/MWh, inelastic demand. α = 0.9, β = γ = 1, ε = ζ = 1, $\bar{s}$ = 800 MWh, w = 40.

- The battery charges when renewable energy is in surplus (t = 1, 3, 4) and discharges instead of thermal (t = 2, 5, 6).
- It is empty at t = 2 and 6 and full at t = 4.

Marginal values (Prop. 4)

$$v_1 = \frac{43.1+1}{0.9} = 49 \quad \text{charge at } t = 1$$
$$v_2 = \frac{0+1}{0.9} = 1.11 \quad \text{charge at } t = 3$$
$$v_3 = \frac{50-1}{1} = 49 \quad \text{discharge at } t = 5$$
$$w = 40 \quad \text{last segment}$$

π and rents (Cor. 4)

$$\pi = (49,\ 1.11,\ 1.11,\ 49,\ 49,\ 40)$$
$$r_2 = 47.89, \quad r_4 = -47.89, \quad r_6 = 9$$

Equal to the multipliers returned by the solver

NEW FINDING

Prices pin the value of stored energy almost everywhere, except where the battery hits a limit it cannot undo

2 of 6

hours with a free π in the base case (t = 2, 4)

[1.11, 49]

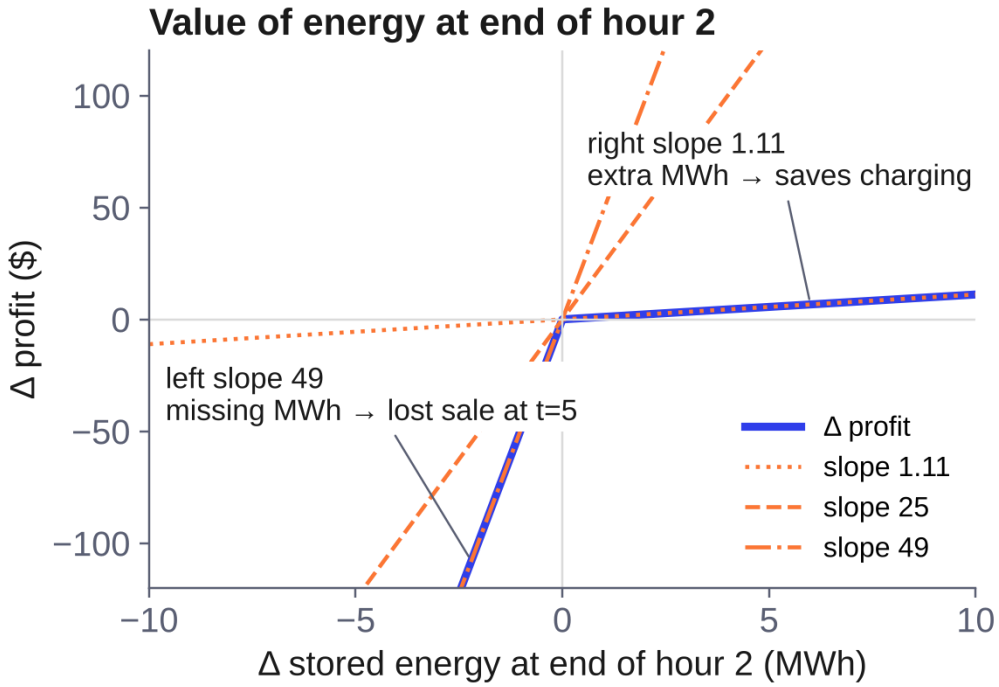
admissible range of π_2 and π_4 ($/MWh)

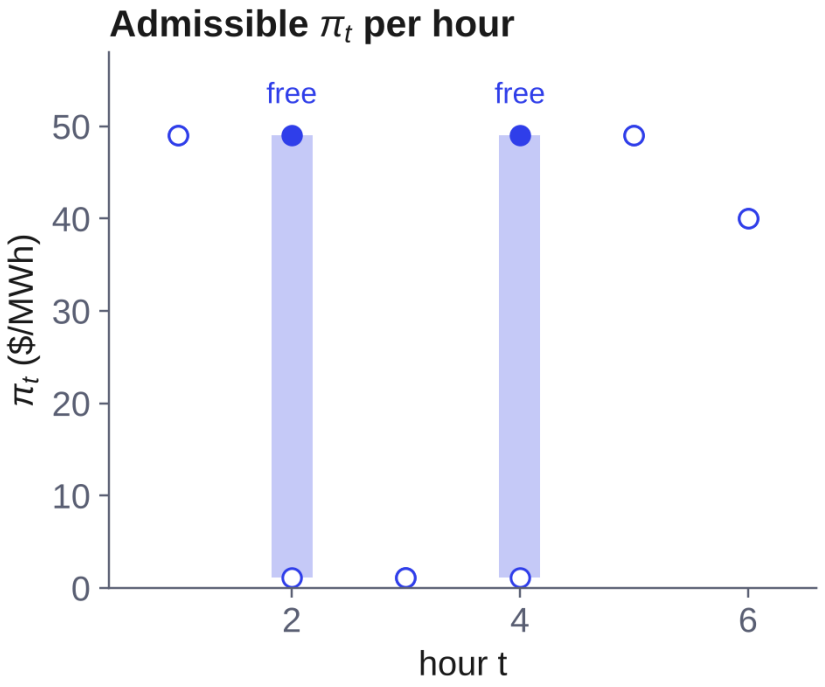
same λ★

for every admissible choice, with perfect forecasts

Why two hours are free: a kink in the value of energy

NON-UNIQUENESS





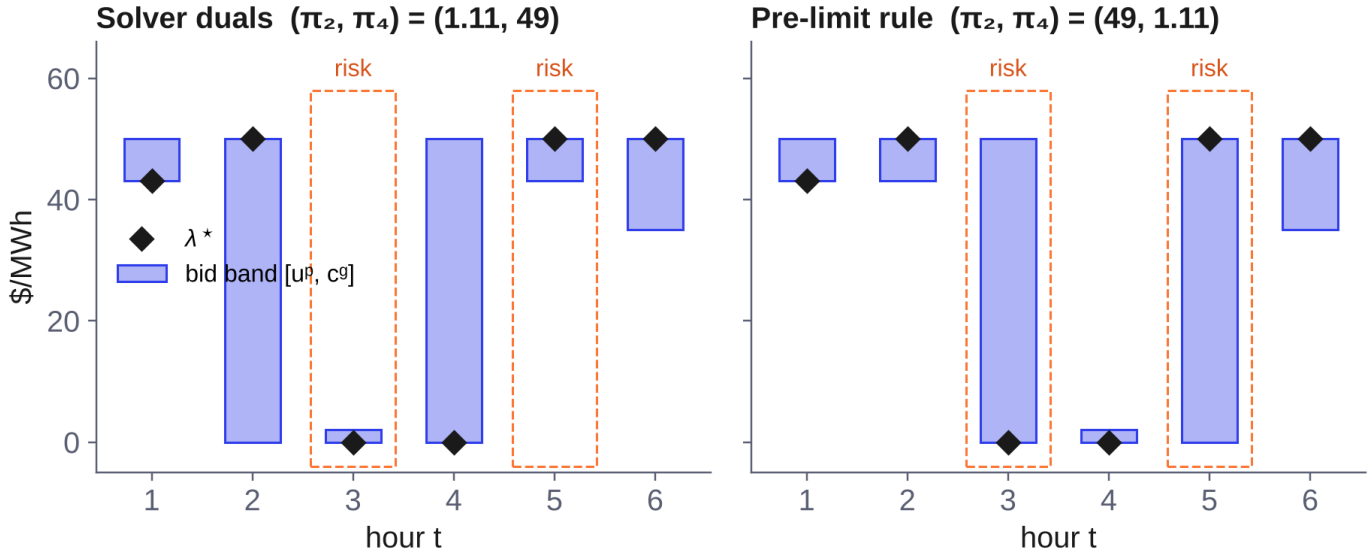
$$\pi_2 \in \Big[\underbrace{\frac{\epsilon}{\alpha} = 1.11}_{\text{extra MWh: saves charging}} , \underbrace{\frac{\lambda_5 - \zeta}{\beta} = 49}_{\text{missing MWh: lost sale at } t=5} \Big]$$

- t = 2: the battery is empty and the renewable surplus at t = 3 is fully used, so a missing MWh cannot be recharged.
- t = 4: same kink, because the cap binds (720 + 3.6·22.2 = 800).
- Checked by re-solving the LP with ±0.5 MWh added to the SOC recursion.

Same prices, different bids: the choice decides which error you risk

IMPLICATION FOR BIDS

Value of stored energy π	t = 1	t = 2	t = 3	t = 4	t = 5	t = 6
Admissible (LP)	49	[1.11, 49]	1.11	[1.11, 49]	49	40
Solver duals	49	**1.11**	1.11	**49**	49	40
Pre-limit rule	49	**49**	1.11	**1.11**	49	40



$$u^p = (43.1,\ 0.9\pi_2 - 1,\ 0,\ 0,\ 0.9\pi_4 - 1,\ 35)$$
$$c^g = (50,\ 50,\ 1 + \pi_2,\ 1 + \pi_4,\ 50,\ 50)$$

Only the band edges at t = 2–5 depend on π_2 and π_4.

How far can the price move before the battery does the wrong thing?

t	Planned → unwanted	Solver	Pre-limit
2	sell → buy	↓ 50	↓ 6.9
3 ⚠	buy → sell while empty	↑ 2.1	↑ 50
4	buy → sell	↑ 50	↑ 2.1
5 ⚠	sell → buy while full	↓ 6.9	↓ 50

↑ / ↓: price rise / drop ($/MWh) from λ★ that triggers the unwanted action.
⚠ impossible action: infeasible schedule. Amber: feasible, but costs welfare.

Takeaway

- Perfect forecast: every choice gives λ★.
- With errors, each choice leaves two hours exposed: one bid per hour cannot depend on the SOC.

Carry the pre-limit value forward

SELECTION RULE

Pre-limit rule

$$\pi^R_{e,t} = \mathrm{Proj}_{[\underline{\pi}_{e,t}, \bar{\pi}_{e,t}]} \left(\frac{\pi^R_{e,t-1}}{\gamma_e} \right)$$

- The projection interval is the admissible range of π at t: its ends are the right and left slopes of the optimal value.
- Free hours get zero rent: the rent moves to the next hour.

Base case

$$\pi^R = (49,\ 49,\ 1.11,\ 1.11,\ 49,\ 40)$$
$$u^p = (43.1,\ 43.1,\ 0,\ 0,\ 0,\ 35)$$
$$c^g = (50,\ 50,\ 50,\ 2.11,\ 50,\ 50)$$

What it buys

- Margin 50 $/MWh at the infeasibility-risk hours (t = 3, 5).
- Prop. 4 still holds: H = {3, 5, 6}, segments {1,2}, {3,4}, {5}, {6}.

What it costs

- Small margins move to t = 2, 4 (6.9 and 2.1), where a wrong action is feasible and only costs welfare.
- A candidate rule, to be tested in Stage 1.

How an ESS builds its bids without duals

IN PRACTICE · BID FAMILY B^{FCST}

INPUT

Price forecast

own nodal price, T intervals

→

STEP A

Self-schedule LP

against the forecast: charge, discharge, SOC (primal only)

→

STEP B

Read the SOC path

limit-hit hours, marginal intervals

→

STEP C

Values v and π

Prop. 4 + pre-limit rule

→

OUTPUT

Bids u^p, c^g

blocks $\underline{p}$, $\bar{g}$ → submit to DAM

Perfect forecast

- Reproduces Baldick's nodal prices for every admissible π (difference ≈ 1e−15).
- Quantities can differ at the bid edges, so a tie-break rule is needed.

Forecast errors

- Welfare gap ΔSW > 0 with respect to Baldick's design.
- Possible SOC-infeasible schedules, corrected in real time.
- Measured in Stages 1–2.

B^{fcst} only uses information the ESS has: its own data and a forecast of its own nodal price.

NEXT STEPS

Open points before Stage 1

1. Tie-break rule for quantities: at λ★ the current design can clear a different, lower-welfare schedule
2. Price-taker assumption: for a large ESS, the price forecast depends on its own bids (fixed point)
3. Nodal forecasts: with congestion, the forecast must be the ESS's own LMP
4. Stage 1: implement the forecast-based bids with the pre-limit rule and test it with forecast errors (ΔSW, infeasible hours)

Reference: R. Baldick, "A state-of-charge based formulation for storage participation in electricity markets," arXiv:2606.14993, 2026.


DTU